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"What Your Customers Want"

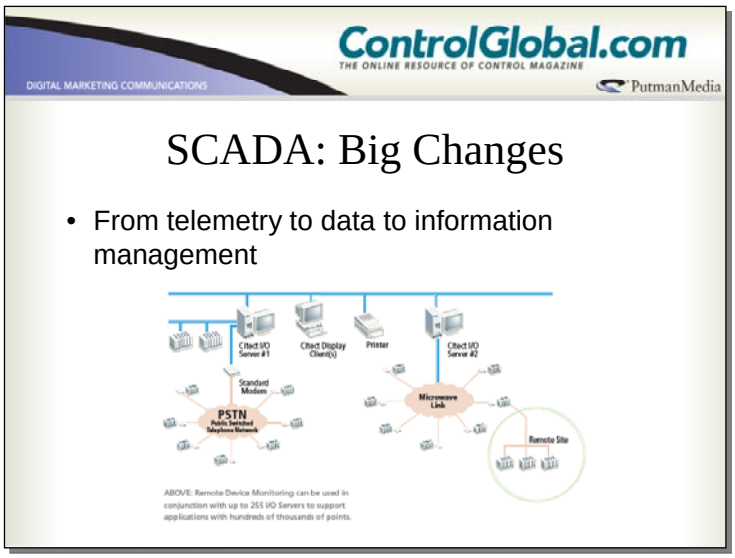
by

Walt Boyes
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My name is Walt Boyes, and I am the editor of Control magazine. But I have been an end-user, a field engineer, a system designer, a field salesman, a sales manager, a marketing manager, and I've run a couple of companies, including a manufacturers' rep company. That's by way of letting you know that I know who you are, and what makes your motors run. There was, as you know, a fancier title for this talk, but fundamentally, I'm going to be talking about what your customers want.



When I did my first radio SCADA surveys in 1976, and installed my first telemetry and control systems, SCADA wasn't anything like it has become. In the past thirty years, we've gone from simple FSK tone telemetry and leased telephone lines to systems that are high speed Ethernet on fiber-optic backbones, or microwave, radio, and satellite transmission systems carrying huge volumes of data.

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We've gone from just barely able to get alarms and maybe if we were lucky, pump run times, back to the SCADA head end– to being able to control whole systems completely remotely from thousands of miles away.

And now, we aren't just looking for alarms anymore.

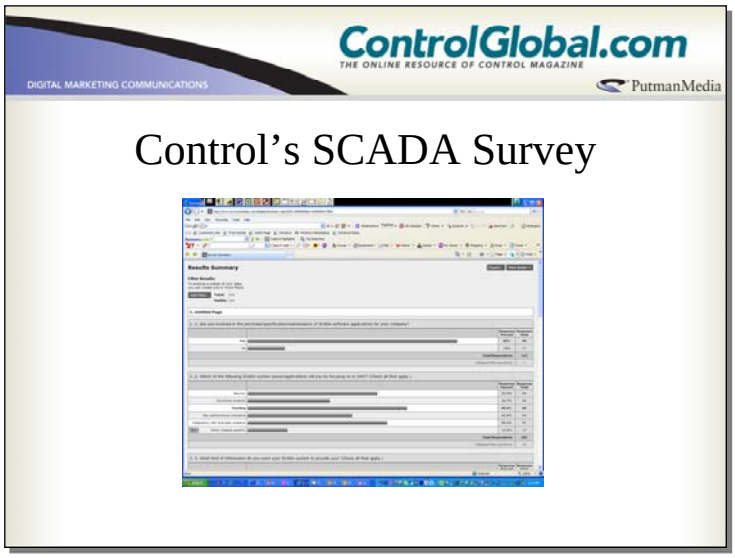
Now we're looking to integrate the data brought back from each remote station and each field device and final control element into asset management, MES and ERP systems.

It is pretty amazing what thirty years will bring.



Like you, I've been intimately involved in the automation industry for most of my life. My father began in the industry in 1940 when he joined a little company called Brown Instrument– now better known as Honeywell. When I was growing up, he was a manufacturers' rep in the water and wastewater industry, and in 1975, I joined and then led the company for eight years. One thing I learned from being on the front lines in the SCADA industry is to listen, very close and very hard, to my customers. So what I want to do is to share with you some things we've heard from your customers.

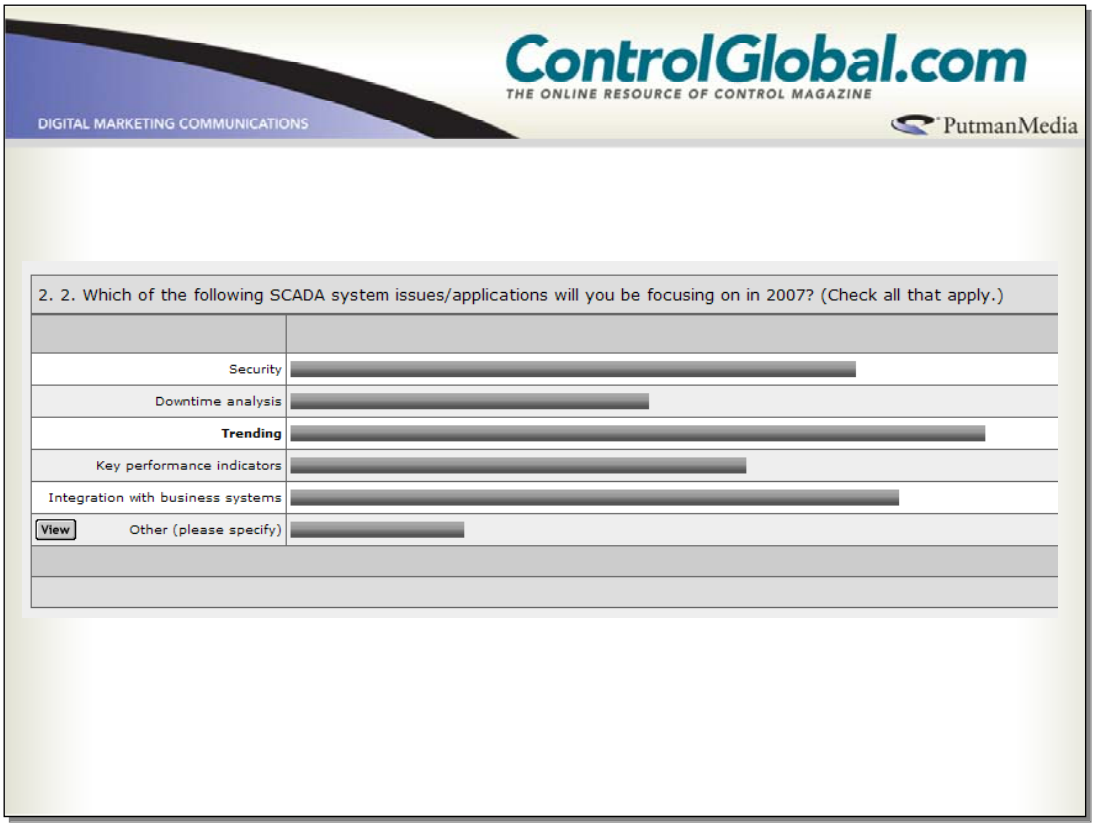
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Last month, we surveyed over 100 end user readers who are users of SCADA systems. We asked them questions about what they want and what they need both in the strategic future and in the immediate future from their SCADA suppliers.

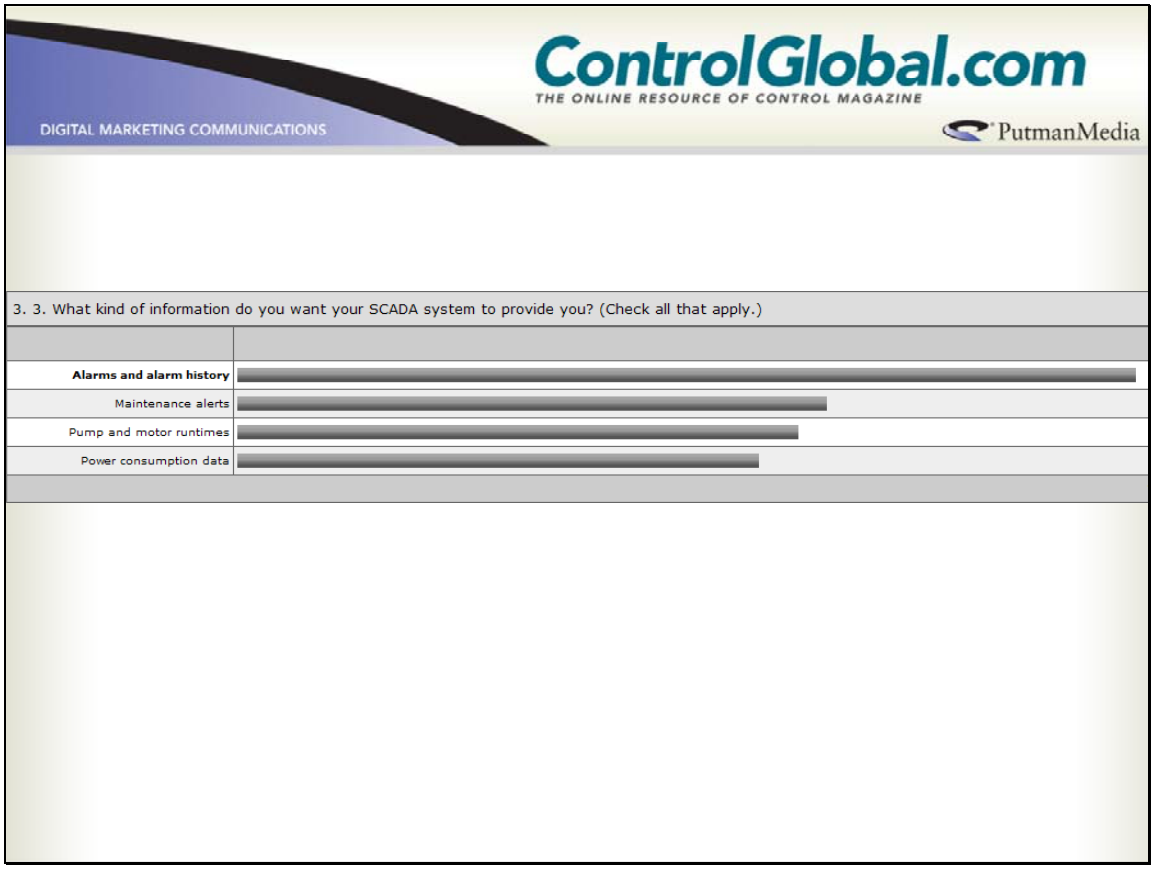
Let's take a look at the data.

The first question we asked was "Are you involved in the purchase/specification/maintenance of SCADA software applications for your company?" 85% of the respondents said yes. So we know we got the right



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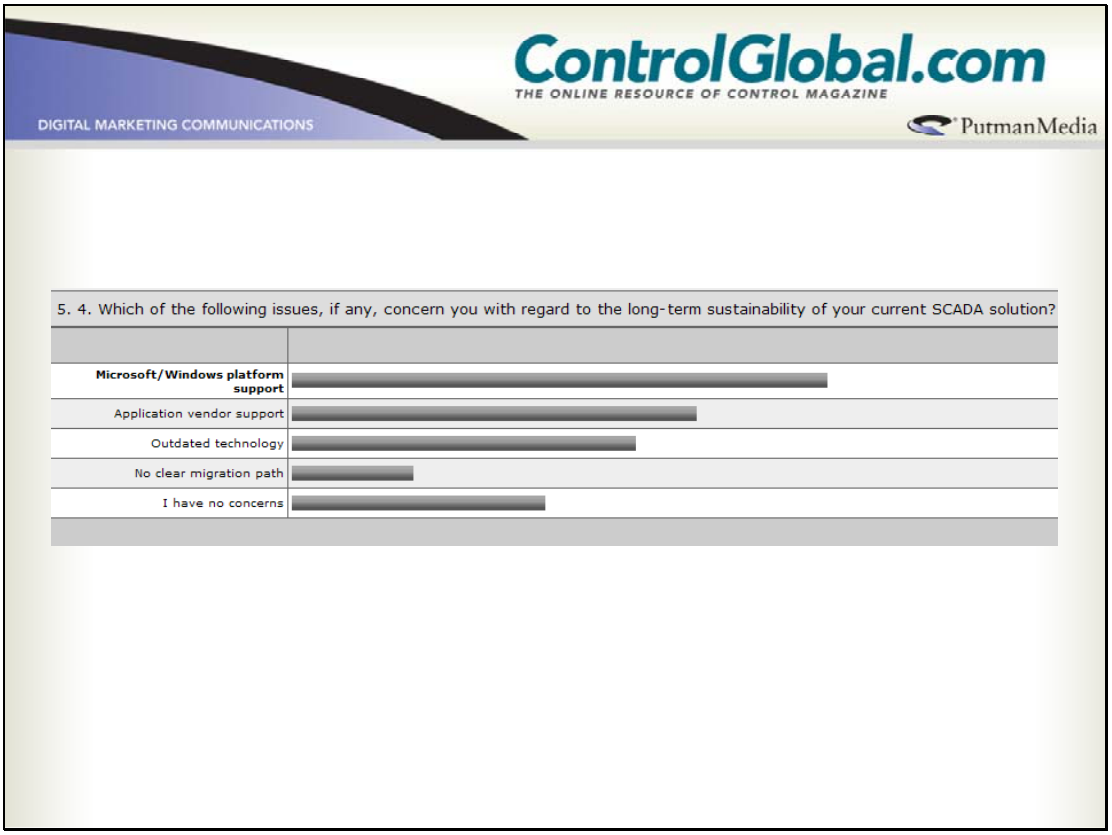
people. 65% of the respondents said that trending would be a major focus, with over 50% checking security. The second most checked item, you'll note, was integration with business systems. A significant minority is interested in asset management. The "other" category split 14 different ways, including some answers that probably could be grouped under the major headings.



By a wide margin, almost 94%, your customers are most interested in alarms and alarm history– just like SCADA users have been for over 30 years. They also want maintenance alerts, runtimes and power consumption data. The more things change, the more they stay the same.

In addition, they wanted things like weather, engineering analysis for planning, equipment health, OEE and productivity metrics. So your customers aren't hidebound either.

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There were some interesting answers here. 53% of the respondents were focused on Microsoft when thinking about long-term sustainability of their existing system. Next, however, was application vendor support, followed by concerns over outdated technology.

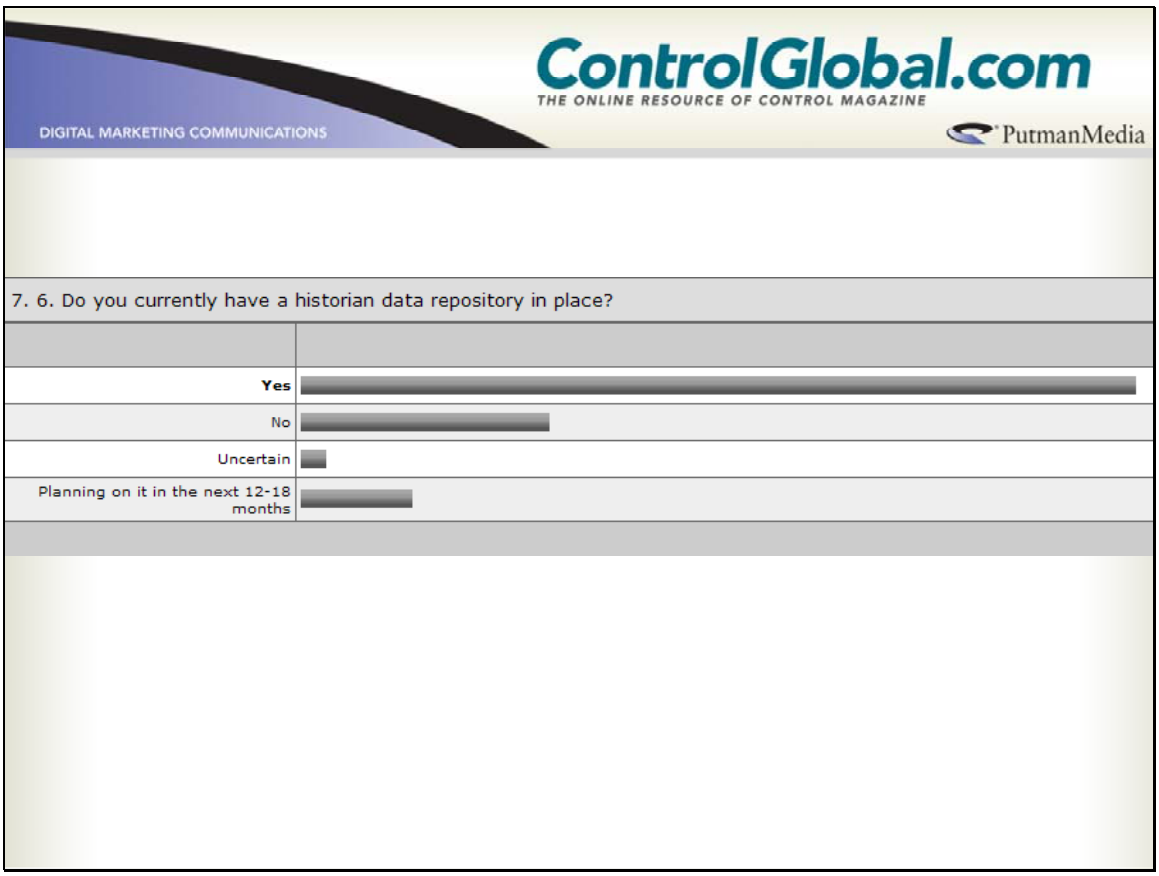
Of course there were those poor, simple souls who had no concerns. I wonder what they really meant by that.

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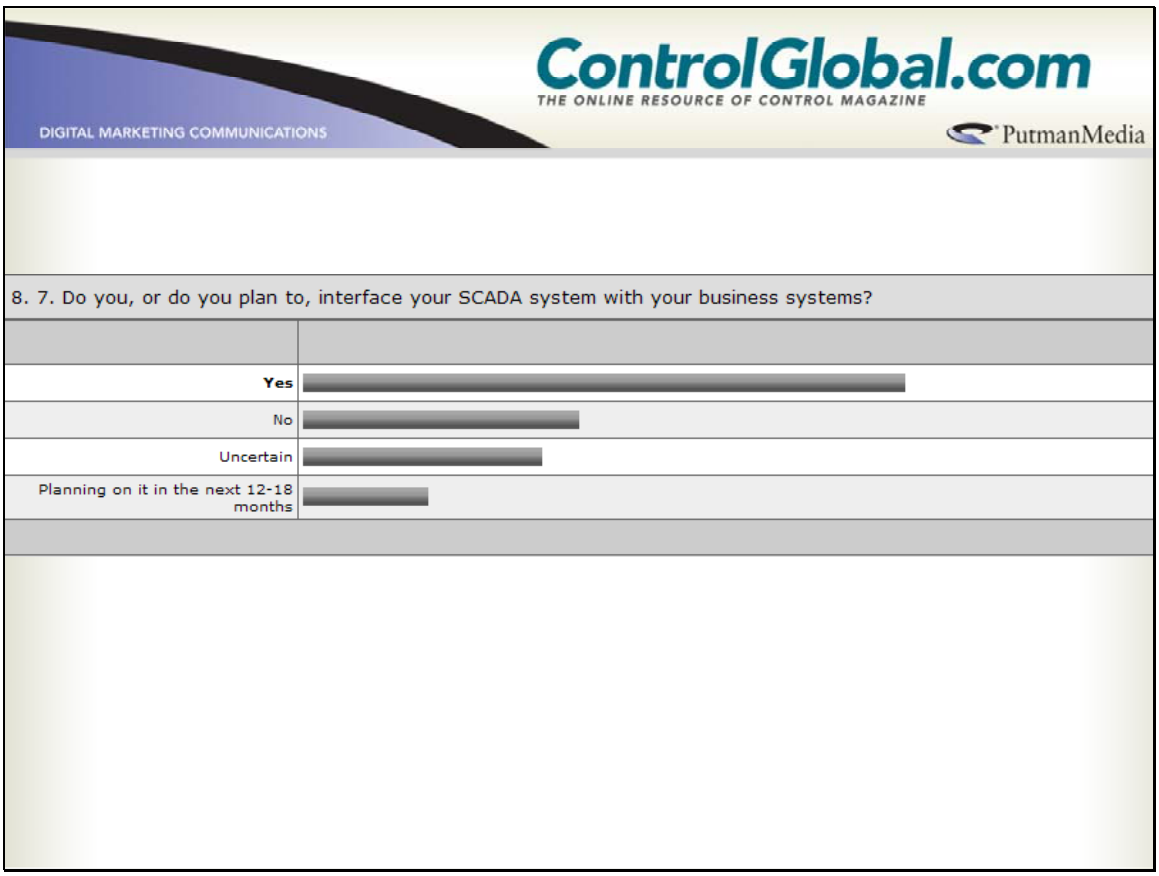
Most (well over 50%) of the respondents are applying security updates. But a significant number either have found out they're having trouble or aren't sure– a sure sign they're having trouble. Interestingly, only 5% indicated they were going to address the issue in the next 12-18 months.

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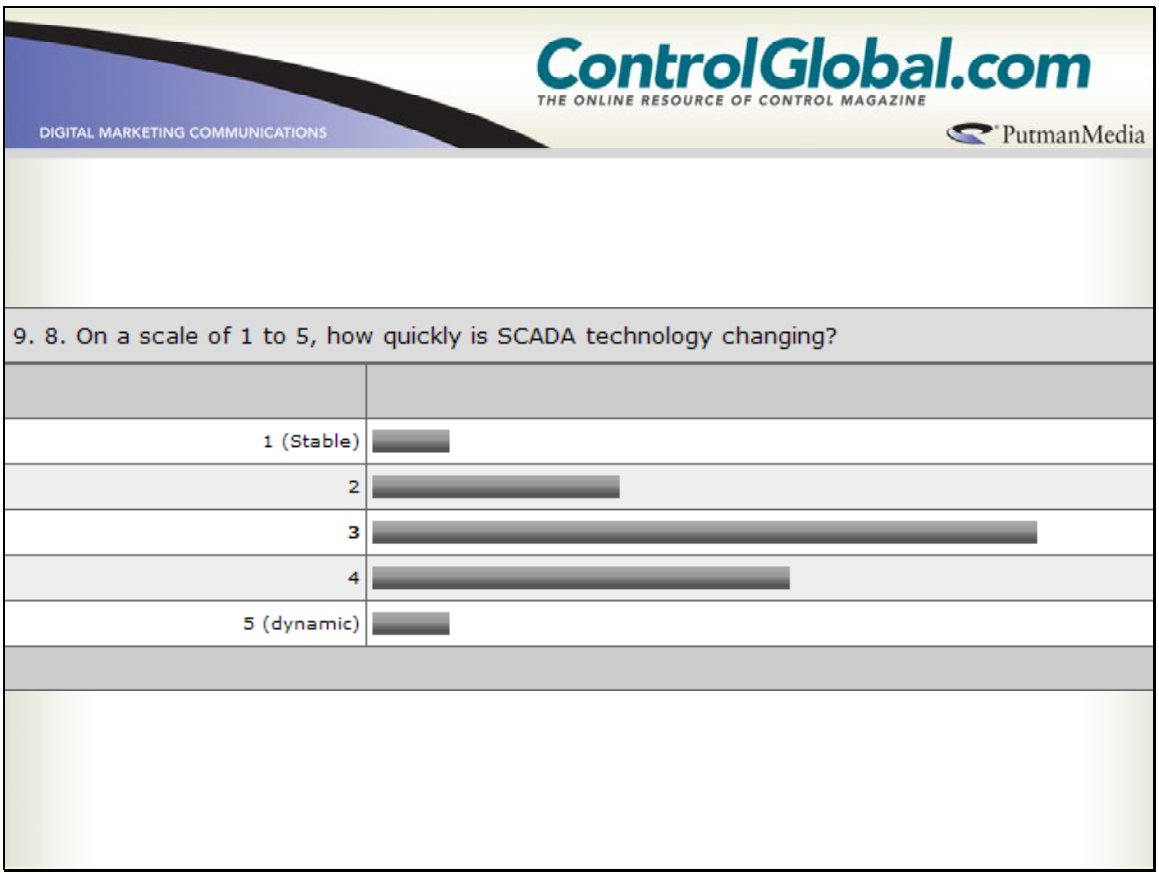
Most people admitted to having some sort of data historian in place. But don't be concerned…most, in this case, only means 67%. A good 10% plan to add one in the next near term. That's good news if you have one to sell.

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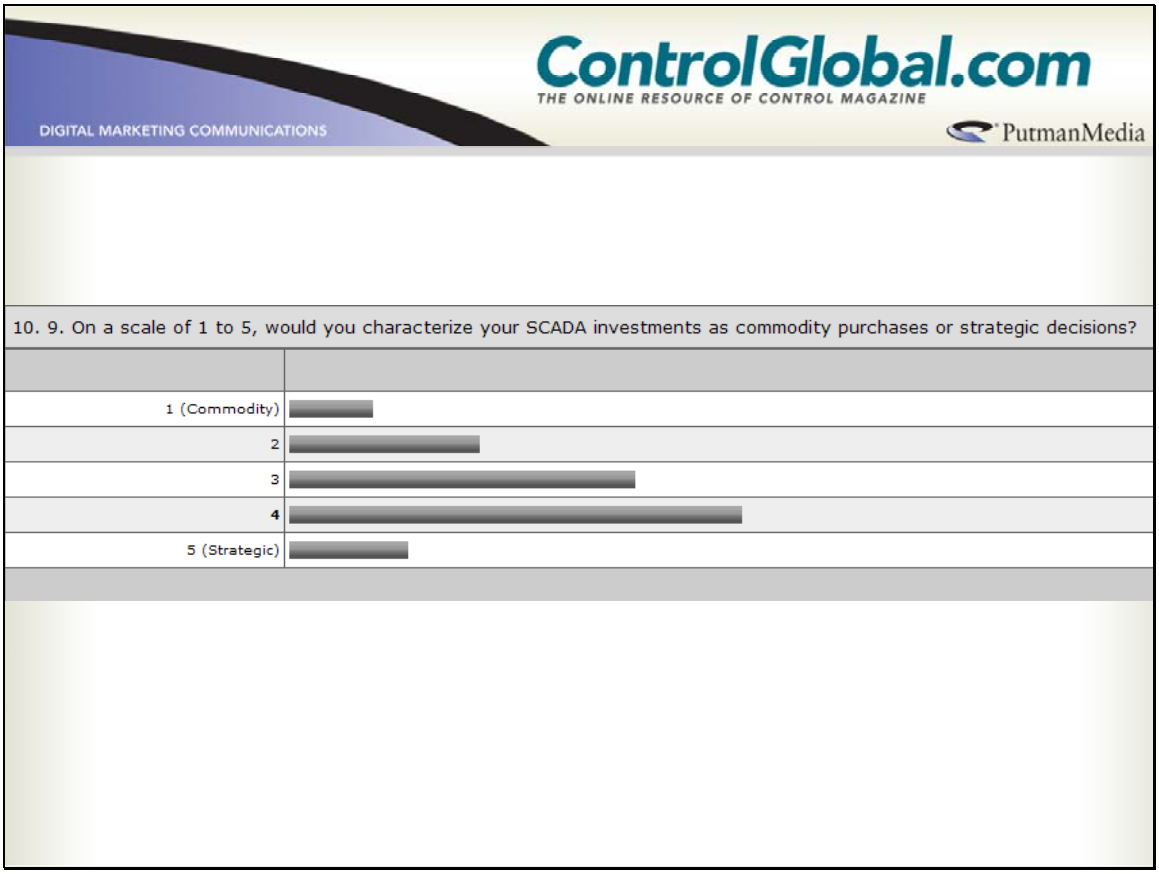
Now look at this. About 50% of respondents say they plan to interface their SCADA system with the enterprise systems they also run. Only 22% said no, and again, 10% say they're in the market to buy in the near term. Note that nearly 20% of respondents are uncertain if they will do it or not. That's an education campaign waiting to happen.

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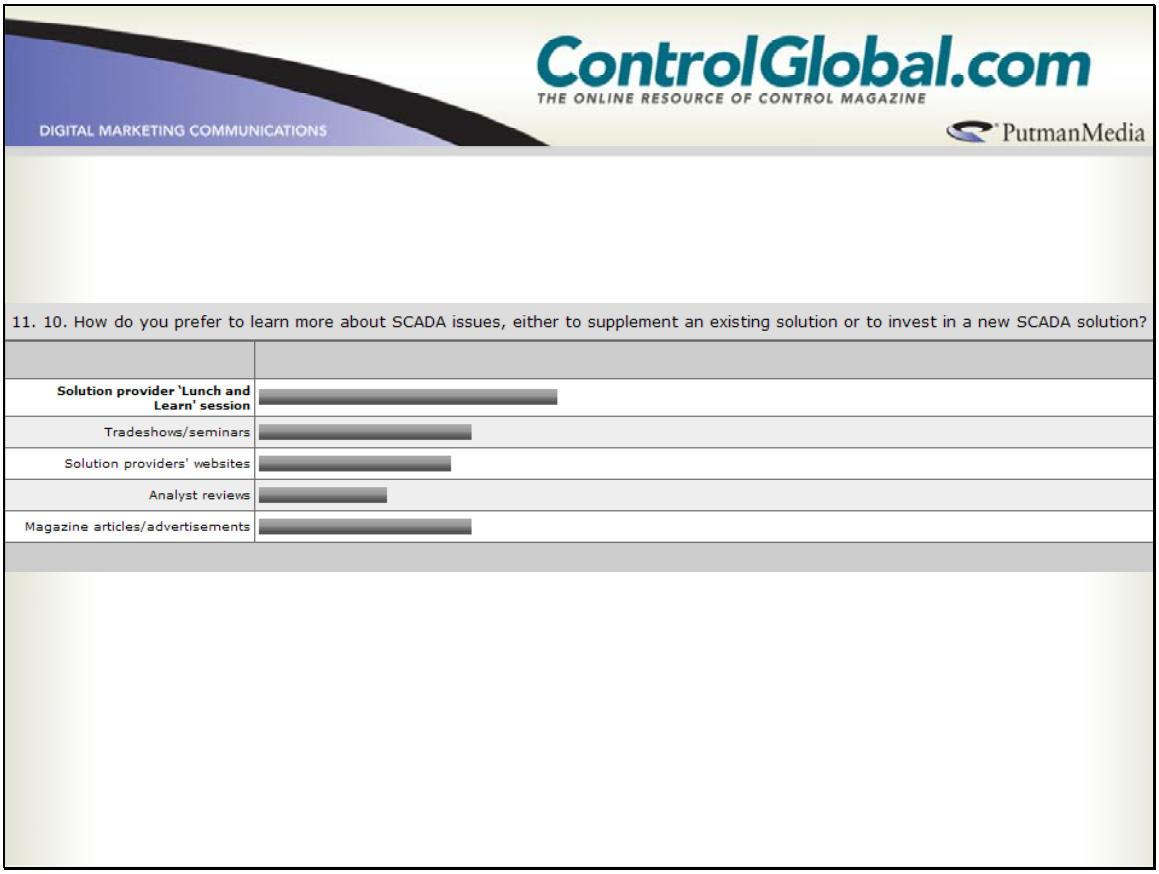
My thesis is that SCADA is changing rapidly. The end users agree. 75% chose a response between 3 and 5, while only 6% believe that the technology is stable. The technology is in fact highly volatile, with many challenges and changes coming in the next few years.

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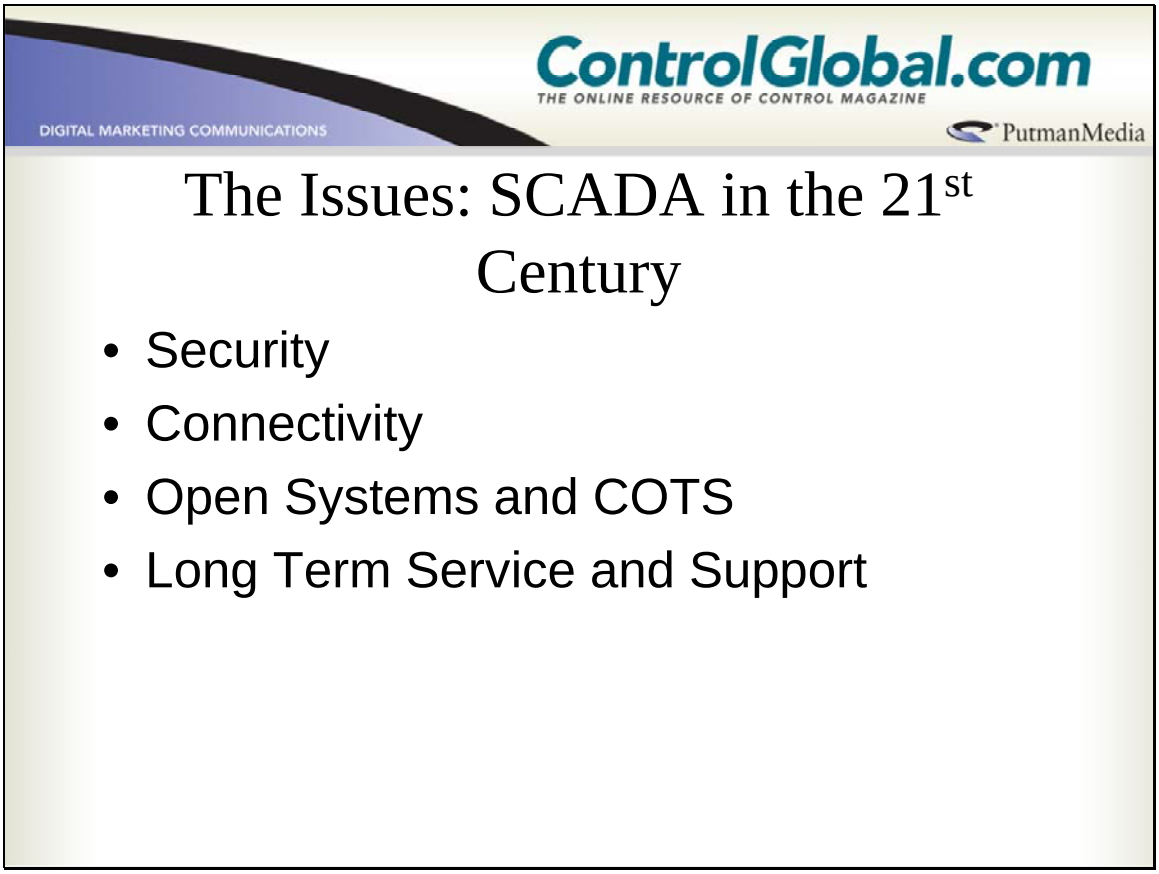
End users understand clearly that a SCADA system is a strategic purchase, not a commodity. Fewer than 7% chose 1, while nearly 50% chose 4 and 5. When you add in the choosers of the middle value, 3, you get 77% of respondents who believe in the strategic nature of the purchase of SCADA systems. What this means is that they're interested in something more than just low price. Clearly, from the spread you can see here, price IS an object…but it isn't by any means the most important part of the decision process.

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The survey replicated attitudes we've seen before. At ISA's first annual Marketing and Sales Summit Conference last September, Honeywell's Brian Chapman presented similar results from studies he's commissioned. Unlike the other parts of the high tech industries, computers, telecoms, and networking, automation professionals have a higher trust relationship with suppliers than with other information providers. In order, it is suppliers, magazines, and then the analyst community. This is different than the rest of the high tech industries, as I said, where the order is magazines, analysts, and then the suppliers. Fact is, end users want relationships with people who know what they are talking about, and people they can trust to be there when it all starts to go down the ubiquitous porcelain convenience. We've all been there, haven't we?

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As we've seen over the past few years, the security of control systems is becoming a serious matter of debate. In fact, if you've been following it, there has been a lively debate going on on the SCADA list for several weeks concerning security, with highly respected practitioners taking positions as varied as "It'll all sort out" to "There ought to be no connection to anything from or to the SCADA system."

As wireless sensors and transmission become more common, we can expect to see more unforeseen vulnerabilities. Here's one from the DCS world, but it could, with just a little tweaking, work just fine in the SCADA world….

Somebody of malicious intent gets a job working in a remote engineering facility that does VPN based support for a control system vendor. Perhaps this facility is located in a place in the world where there are unfriendly people. The hacker

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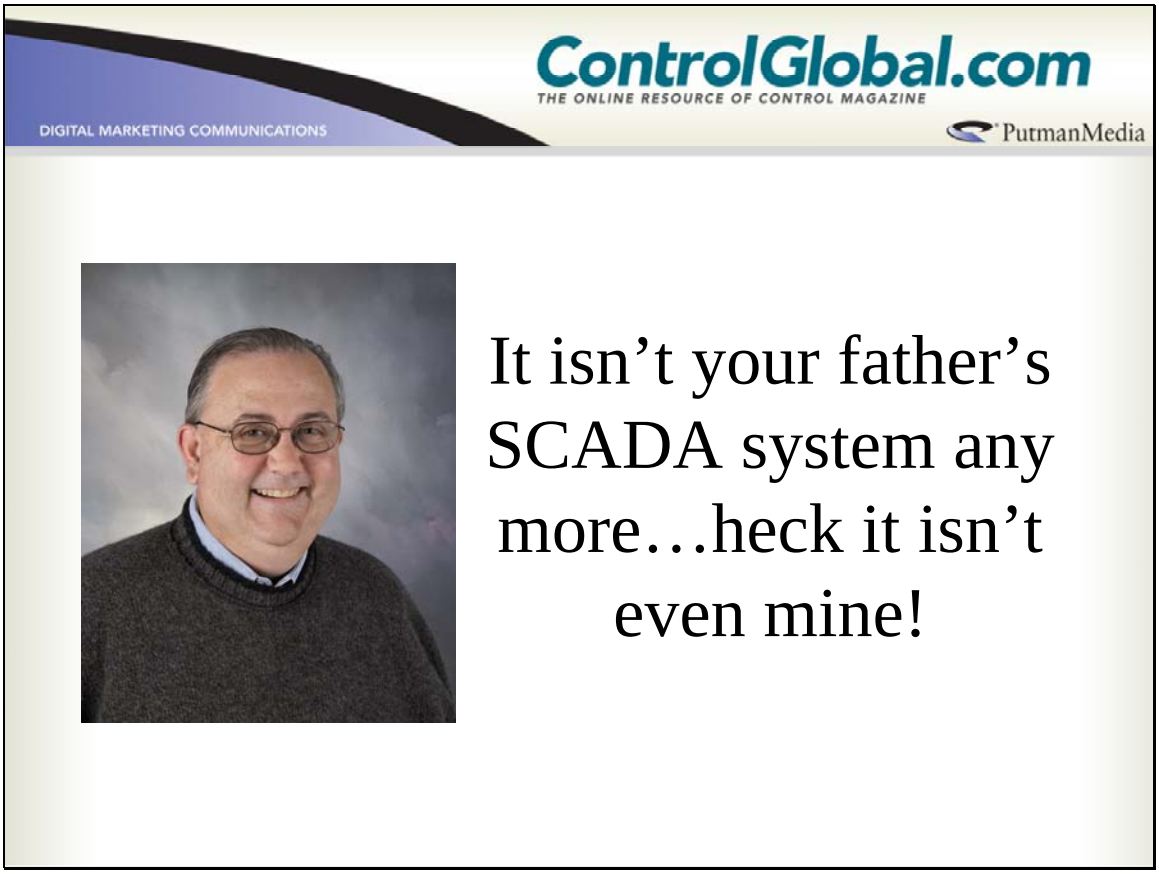
gets access, by virtue of his job, to your control system. You run a refinery. He uses the fact that in order to provide asset management services to you, he has almost complete trusted access to your control system. He first disables the Safety Instrumented System, and then spoofs the HMI so that the operators are completely unaware there is a problem brewing. Finally, he brings the plant down. This is a real nightmare, that keeps SCADA and process control security experts up nights.

SCADA systems, especially, were and are designed to be easy to interface to…using OPC, XML, and other open system tools…and now we have to make sure that only people we trust can use them.

This is especially true since those very same end users who are now concerned about security have been pushing the use of open systems, and Commercial Off The Shelf solutions, for the past fifteen years. No, they don't want a proprietary console, and a proprietary network architecture and encryption scheme. They want easy to establish connectivity between suppliers, and between databases, and they don't intend to give that up.

One of the things I find interesting is that the issue of long term service and support is divided. You see, there are lots of people like the Greater Vancouver Regional District in British Columbia, who have systems that are years old, but which are working just fine, thank you very much, and they don't want to have to replace them if they can possibly avoid it. So, they want support for those older systems, and clear and supported upgrade paths to new systems when they finally can't buy parts on eBay anymore, and the last person who wrote code has retired.

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Be prepared to see the rate of change in SCADA technology continue to increase. We'll see better technologies, newer ways of handling data, and better control systems in the next few years. In thirty years, the products we think are state of the art now will be as antiquated looking as a single FSK tone module is today. As Jeff Harrow, one of my columnists, and a well-known futurist, says, "Don't Blink!"

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